

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Easy

Time allowed: 20

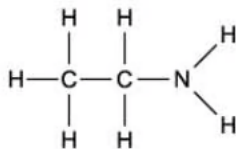
Score: /10

Percentage: /100

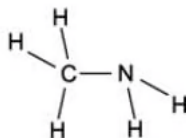
Question 1

Which of the following is the displayed formula for ethylamine?

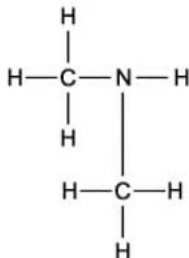
A.



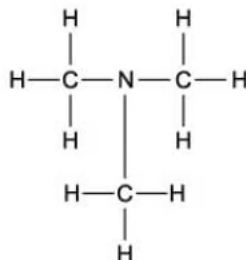
B.



C.



D.



[1 mark]

Question 2

What type of reaction makes ethylamine from bromoethane?

- A. Oxidation
- B. Reduction
- C. Nucleophilic substitution
- D. Electrophilic substitution

[1 mark]

Question 3

$\text{CH}_3\text{CH}_2\text{Cl}$ reacts with an excess of ethanolic NH_3 .

Which compound is the main organic product?

- A. $(\text{CH}_3\text{CH}_2)_3\text{N}$
- B. $(\text{CH}_3\text{CH}_2)_2\text{NH}$
- C. $\text{CH}_3\text{CH}_2\text{NH}_2$
- D. $(\text{CH}_3\text{CH}_2)_4\text{N}^+$

[1 mark]

Question 4

Hydrolysis of $\text{CH}_3\text{CH}_2\text{CN}$ with dilute HCl produces which compound?

- A. CH_3COOH
- B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_3$
- C. $\text{CH}_3\text{CH}_2\text{COOH}$
- D. $\text{CH}_3\text{CH}_2\text{COH}$

[1 mark]

Question 5

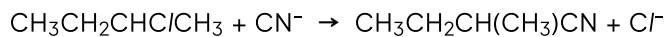
When 1-bromoethane reacts with an equal amount of **X**. What would **X** be and what is the product of this reaction?

	X	Product
A	Ammonia	Amine
B	Methylamine	Amide
C	Water	Weak acid
D	Methanol	Ethyl methanoate

[1 mark]

Question 6

The equation shows a chemical reaction



What is the name for this type of reaction?

- A. Nucleophilic substitution
- B. Electrophilic substitution
- C. Free radical substitution
- D. Nucleophilic addition

[1 mark]

Question 7

What are the reagents needed in the reaction of 1-bromopropane to butanenitrile?

- A. KCN
- B. Ethanol and heat
- C. KCN, ethanol and heat
- D. KCN and ethanol

[1 mark]

Question 8

What are the products in the reaction of ethanenitrile and hydrochloric acid?

- A. Propanoic acid and ammonia
- B. Propanoic acid and ammonium chloride
- C. Ethanoic acid and ammonia
- D. Ethanoic acid and ammonium chloride

[1 mark]

Question 9

What are the products in the reaction of ethanenitrile and sodium hydroxide?

- A. Sodium propanoate and ammonia
- B. Sodium propanoate ammonium chloride
- C. Sodium ethanoate and ammonia
- D. Sodium ethanoate and ammonium chloride

[1 mark]

Question 10

Which one of the following reactants would **not** produce a nitrile in a one-step reaction?

- A. Amides
- B. Aldehydes
- C. Halogenoalkanes
- D. Alkanes

[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- 'Ethyl' suggests that the carbon chain consists of two carbon atoms.
- '-amine' suggests that the -NH_2 group is located at the end of the alkyl chain.
- Therefore, compound A is the primary amine **ethylamine**.

B is incorrect as this compound is **methylanine** and consists of only one carbon atom instead of two.

C is incorrect as this compound is a secondary amine **dimethylanine** (two methyl groups attached to an amine group).

D is incorrect as this compound is a tertiary amine **trimethyl amine** consisting of three carbon atoms instead of two.

2

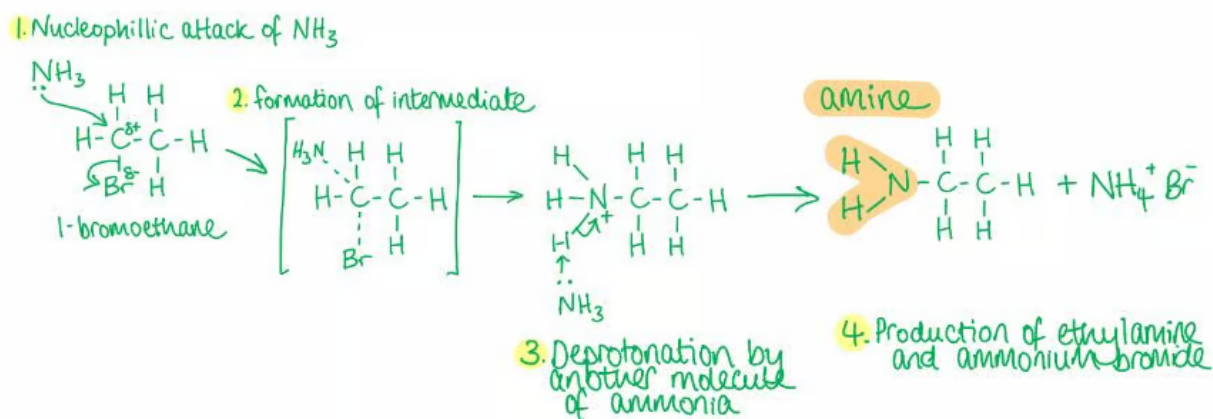
The correct answer is **C** because:

- In nucleophilic substitution reactions, a nucleophile replaces another nucleophile in the compound.
- A **nucleophile** is molecule which **donates** a pair of electrons.
- Bromoethane undergoes nucleophilic substitution with **ammonia** using **excess hot ethanolic ammonia** under pressure.
 - If ammonia is not used in excess, a mixture of secondary and tertiary amines can be produced.
- In this reaction, **ammonia** acts as the **nucleophile** and replaces the **bromine** in to form **ethylamine**.
- Bromoethane is a **primary halogenoalkane** so the nucleophilic substitution of

bromoethane by ammonia follows an S_N2 reaction mechanism.

- Primary halogenoalkanes undergo **S_N2** substitution reactions.
- Tertiary halogenoalkanes undergo **S_N1** substitution reactions.
- Secondary halogenoalkanes undergo a **mixture** of S_N2 and S_N1 substitution reactions depending on their structure.

- The reaction mechanism for this reaction is as follows:



A is incorrect as oxidation of bromoethane produces **ethanol** (an **alcohol**).

B is incorrect as reduction of bromoethane produces **ethene** (an **alkene**).

D is incorrect as ammonia (the reagent) is **nucleophile** not an electrophile.

3

The correct answer is **C** because:

- The reaction of a halogenoalkane with **excess hot ethanolic ammonia** under pressure results in the formation of a **primary amine**.
- This is a **nucleophilic substitution** reaction in which the ammonia acts as the nucleophile and replaces the chlorine atom in **chloroethane** to form **ethylamine**.
- Compound C is the only primary amine.
- A mixture of amines will be formed when ammonia is **not** used in excess.

A, B & D are incorrect as these are **not primary amines**. A is a tertiary amine, B is a secondary amine and D is a quaternary ammonia ion formed from the **nucleophilic substitution** reaction of chloroethane with ammonia when ammonia is **not** in excess.

4

The correct answer is **C** because:

- **Hydrolysis** is the breakdown of a compound using water.
- When a nitrile is **refluxed** with dilute hydrochloric acid, **hydrolysis** occurs.
- The -CN group at the end of the hydrocarbon chain is converted into a -COOH group, forming a **carboxylic acid**.
- The ammonia in solution will act as a **base** and form an **ammonium salt** with Cl^- (**NH_4Cl**).
- The reaction equation for this **hydrolysis** is:
 - $\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$

A is incorrect as though a carboxylic acid is formed, the carbon chain is one carbon too short.

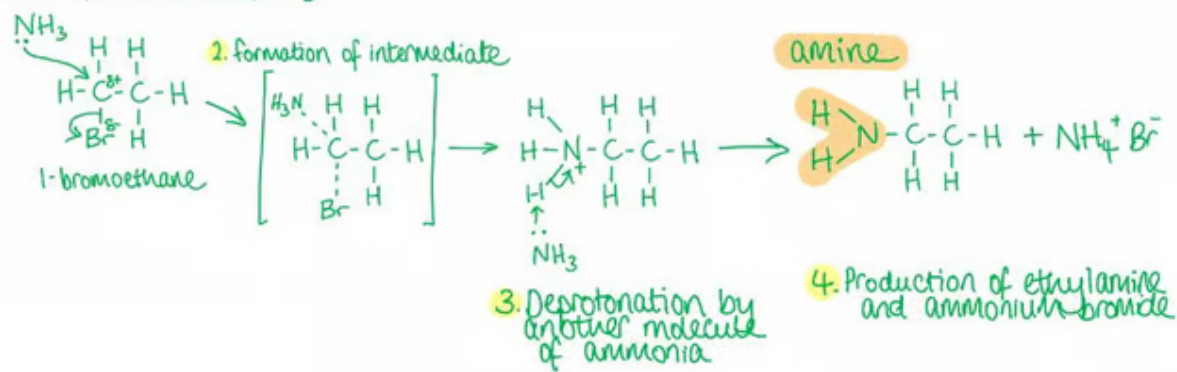
B is incorrect as this is a primary amine which would have been formed if the nitrile group in $\text{CH}_3\text{CH}_2\text{CN}$ was **reduced** instead of **hydrolysed**.

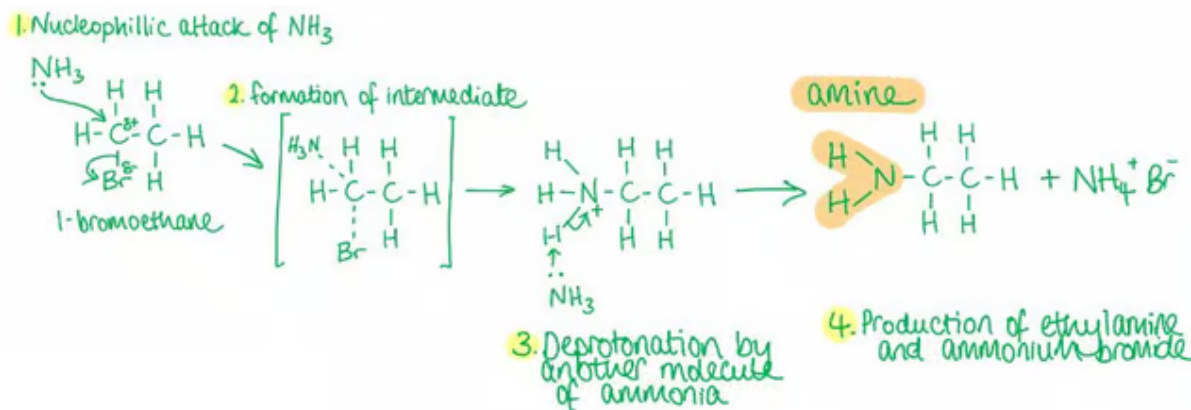
D is incorrect as this is an **aldehyde** and wouldn't form upon **hydrolysis** of the nitrile.

5

The correct answer is **A** because:

- Halogenoalkanes will react with **ethanolic ammonia** to form **amines** in a nucleophilic substitution reaction.
- A **nucleophile** is molecule which can donate a pair of electrons.
- The **ammonia** acts as a **nucleophile** and will replace the bromine atom in 1-bromoethane to form **ethylamine**.
- Bromoethane is a **primary halogenoalkane** so the nucleophilic substitution of bromoethane by ammonia follows an **$\text{S}_{\text{N}}2$ reaction** mechanism.

1. Nucleophilic attack of NH_3 



B is incorrect as reacting 1-bromoethane with methylamine is a **nucleophilic substitution** reaction but will form the **secondary amine** *N*-methylethanamine and not an amide.

C is incorrect as the reaction of halogenoalkanes with water is a **nucleophilic substitution** reaction (also called a **hydrolysis** reaction) to form **alcohols**. 1-bromoethane will therefore get hydrolysed by water to form ethanol which is a **weak base** and not a weak acid.

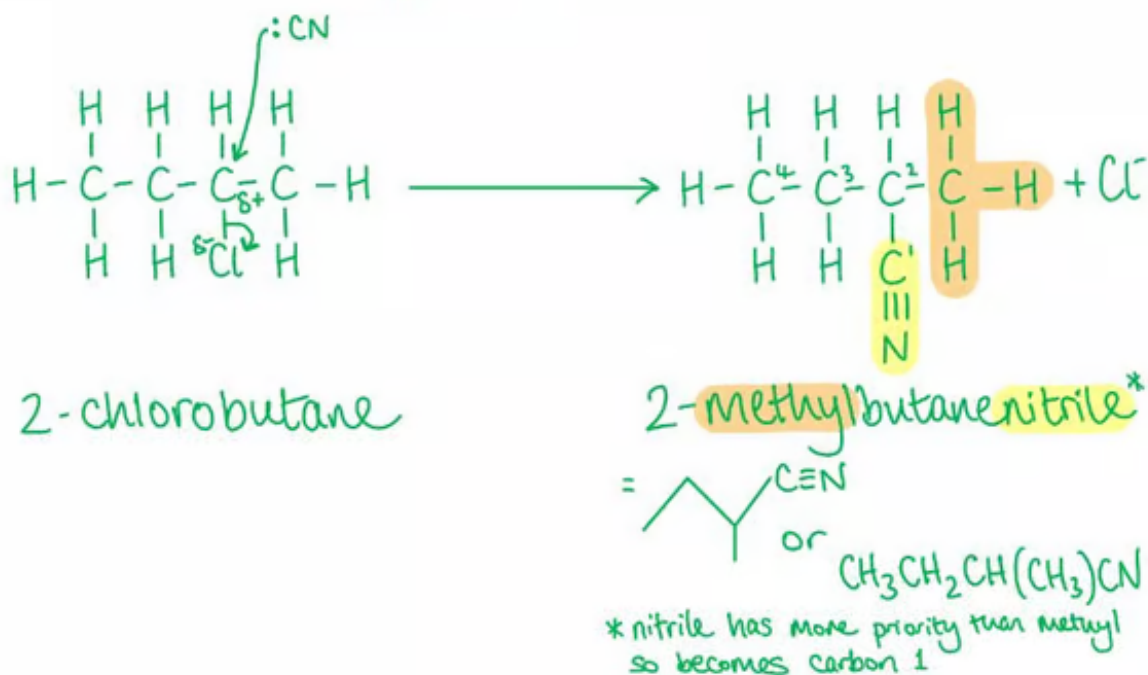
D is incorrect as the reaction of halogenoalkanes with **aqueous alkali** is a **nucleophilic substitution** reaction, therefore, will form **methoxyethane** which is an **ether** and not **ethyl methanoate** which is an **ester**.

6

The correct answer is **A** because:

- A **nucleophilic substitution** reaction is one in which a **nucleophile** replaces another nucleophile.
 - A **nucleophile** is a species which can donate a pair of electrons.
- The cyanide ion (CN^-) in this reaction is the **nucleophile** and replaces the **chlorine atom** in $\text{CH}_3\text{CH}_2\text{CHCl}/\text{CH}_3$.
- 2-chlorobutane is a **secondary halogenoalkane** so the nucleophilic substitution of 2-chlorobutane by ammonia can follow an $\text{S}_{\text{N}}2$ or $\text{S}_{\text{N}}1$ reaction mechanism.
- The reaction of the halogenoalkane with nitriles (CN) will add an extra carbon atom to the carbon chain.
- The reaction mechanism for this reaction via $\text{S}_{\text{N}}2$ is as follows:

Nucleophilic attack by cyanide ion



B is incorrect as CN^- is not an **electrophile** which is an electron pair acceptor.

C is incorrect as a free radical substitution reaction is the reaction in which **halogen atoms** (not CN^-) replace **hydrogen atoms** in alkanes.

D is incorrect as the reaction is a substitution reaction **not** an addition reaction.

7

The correct answer is **C** because:

- A **nucleophilic substitution** reaction is one in which a **nucleophile** replaces another nucleophile.
 - A **nucleophile** is a species which can **donate** a pair of electrons.
- The reaction of halogenoalkanes with cyanide ions, CN^- , is an example of a **nucleophilic substitution** reaction.
- The cyanide ion (CN^-) in this reaction is the **nucleophile** and **replaces** the **bromine atom** in bromopropane to give butanenitrile.
- The reaction of the halogenoalkane with nitriles will add an extra carbon atom to the carbon chain.

- ‘**–propane**’ changes into ‘**butane**–’ as there are now four carbon atoms in the carbon chain instead of three.
- The reaction should be carried out with KCN (to provide the **nucleophile** which is the cyanide ion) in ethanol (also known as **ethanolic solution of potassium cyanide**) which is then **heated** under reflux with the 1–bromopropane.

8

The correct answer is **D** because:

- **Hydrolysis** is the breakdown of a compound using water.
- When a nitrile is **refluxed** with **dilute hydrochloric acid**, **hydrolysis** occurs.
- The –CN group at the end of the hydrocarbon chain is converted into a –COOH group, forming a **carboxylic acid**.
- The ammonia in solution will act as a **base** and form an **ammonium salt** with C/ (**NH₄Cl**).
- The reaction equation for this **hydrolysis** is:
 - $\text{CH}_3\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{COOH} + \text{NH}_4\text{Cl}$

A & B are incorrect as the number of carbons in the carbon chain remains **unchanged**. The carboxylic acid should therefore be **ethanoic acid** and not propanoic acid.

C is incorrect as the ammonia will react in solution to form an **ammonium salt**.

9

The correct answer is **C** because:

- **Hydrolysis** is the breakdown of a compound using water.
- When a nitrile is **refluxed** with **sodium hydroxide solution**, **hydrolysis** occurs.
- Instead of forming an **ammonium salt** (as seen in HC/reflux reactions), a **sodium salt** is formed, and **ammonia gas** is given off.
- The ammonia is formed from the reaction between **ammonium** (NH₄⁺) ions and **hydroxide** (OH[–]) ions.

- The reaction equation for this hydrolysis equation is:
 - $\text{CH}_3\text{CN} + \text{H}_2\text{O} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{NH}_3$

To form the free **carboxylic acid** the final solution has to be acidified with a strong acid such as dilute **hydrochloric acid (HCl)** such that the **ethanoate** ion in the **sodium ethanoate salt** will react with the hydrogen ions to form **ethanoic acid**. Once reacted, ethanoic acid is a weak acid and holds onto the hydrogen rather than dissociating.

A & B are incorrect as the number of carbons in the carbon chain remain **unchanged**. The **sodium salt** should therefore be **sodium ethanoate** and not sodium propanoate.

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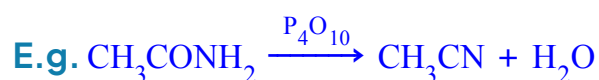
D is incorrect as the ammonium salt is formed when a nitrile is refluxed with dilute hydrochloric acid, **not** sodium hydroxide.

10

The correct answer is **D** because:

- A **dehydration reaction** is a reaction in which a **water molecule** is removed from a compound.
- **Alkanes** are very **unreactive** compounds due to their non-polar C-H bonds.
- They are therefore not likely to undergo reactions with **nucleophiles** and **electrophiles**.
 - Nucleophiles are species that can **donate** a pair of electrons.
 - Electrophiles are species that can **accept** a pair of electrons.
- The conversion of alkanes to nitriles is therefore a **multi-step** process.

A is incorrect as **amides** can undergo a 1-step **dehydration reaction** with **P₄O₁₀** to form **nitriles**:



B is incorrect as **aldehydes** can undergo a 1-step **addition reaction** with **KCN** in **H₂SO₄** at room temperature to form **nitriles**:



C is incorrect as **halogenoalkanes** undergo a 1-step **substitution reaction** with **KCN** in **ethanol** and **heat** to form **nitriles**:

